

KARL HILL

MISSION SOFTWARE

ENGINEERING SYSTEMS | TECHNICAL LEADERSHIP | DELIVERY

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SUMMARY

Staff Aerospace Software Engineer with 20+ years of experience building and delivering production software across national security, aerospace, NASA, and enterprise environments. Combines hands-on engineering with technical leadership across software delivery, engineering practices, mentoring, and cross-team work.

PROFESSIONAL EXPERIENCE

Staff Aerospace Software Engineer

SEPT 2025 — PRESENT

Jacobs National Security · Chantilly, VA

- Lead engineering delivery for a team of about 10 across roughly 20 repositories and multiple deployment environments, sequencing work, coordinating dependencies, and driving integration and release readiness.
- Develop mission software, shared interfaces, messaging integrations, and service orchestration while providing technical guidance and reviewing implementation across the team.
- Advanced a portable messaging layer with a common interface and broker adapters so broker choice can stay in configuration. Ownership is shared.
- Lead Agile planning and execution across internal and partner teams, translating mission priorities into sequenced engineering work and resolving cross-team dependencies and blockers.
- Onboarded and coached approximately six engineers through code review, technical feedback, development guidance, and reinforcement of engineering standards.

Lead Software Engineer

DEC 2017 — SEPT 2025

SSAI / NASA Goddard Space Flight Center · Greenbelt, MD

- Led software engineering on an AWS flood-mapping system for satellite-derived products. GeoHorizons co-author (2026): Software (Equal); Writing – review & editing (Equal). The public map is the shipped artifact.
- Delivered Find Data search, ordering, and near-real-time access for LAADS DAAC, with GitLab CI/CD and Kubernetes web delivery alongside existing archive services.
- Led web engineering on NASA Earth Observatory — editorial workflows, imagery, and the public site. About 1.5 million monthly visitors during that work.
- Automated a content-registry workflow so dataset registration did not depend on a fully manual path.
- Built a Ceph-based file and metadata platform for virtual directory mapping and discovery of large scientific datasets.
- Led Agile software delivery across NASA Earth science teams, coordinating priorities, sprint execution, stakeholder needs, and continuous improvement.

DETAILS

Washington, DC

(202) 599-1442

karlhillx@gmail.com

LINKS

LinkedIn

linkedin.com/in/khill

GitHub

github.com/karlhillx

Website

karlhill.com

AREAS OF EXPERTISE

- Software Engineering
- Technical Leadership & Direction
- Distributed Systems & Integration
- Engineering Systems & Developer Tooling
- Engineer Development & Coaching
- Agile & Cross-Team Delivery

PROFESSIONAL EXPERIENCE

Senior Software Engineer

JAN 2016 — DEC 2017

InformedDNA · Washington, DC

- Designed and delivered a Laravel case-management platform connecting counseling workflows, documentation, and billing.
- Built CRM enhancements for customer lifecycle workflows and reporting, and tightened maintenance and security processes with operations.

Senior Software Engineer

JUN 2012 — MAR 2015

Ticomix, Inc. · Washington, DC

- Delivered SugarCRM solutions for more than 20 clients, including the Virginia Department of Transportation and Kastle Systems.
- Improved backlog management and delivery practices across client projects.

Earlier Engineering Experience

1997 — 2012

Sabre Corporation · Dante Inc. · Visitar Inc. · Verizon Business

- Held software engineering and principal-level roles across travel, enterprise CRM, telecommunications, and managed security.
- At Sabre, built new PHP applications and added features to existing ones for large-scale travel systems.

AREAS OF EXPERTISE

Software Engineering · Technical Leadership & Direction · Distributed Systems & Integration · Engineering Systems & Developer Tooling · Engineer Development & Coaching · Agile & Cross-Team Delivery

PUBLICATIONS

Peer-reviewed research: A web-based high-resolution global water and flood mapping platform. Co-author, A web-based high-resolution global water and flood mapping platform, GeoHorizons (AGU / Geological Society of London), 2026. Software engineering for a NASA-supported system delivering near-real-time, high-resolution satellite-derived water and flood products worldwide. Author contribution (CRediT): Software (Equal); Writing – review & editing (Equal). doi:10.1144/gh2025-7

EDUCATION

Computer Science coursework, University of Maryland, Baltimore County

Associate of Arts, General Studies — Computer Science Curriculum, Howard Community College

Project Management Certificate, Rutgers University

CERTIFICATIONS

SAFe Agilist, Scaled Agile

Professional Scrum Master II, Scrum.org

Professional Scrum Product Owner I, Scrum.org

OPEN SOURCE

bb-run — Run Bitbucket Pipelines locally.

testrisk — Rank the highest-value Python test gaps from coverage, AST, and git.

TECHNICAL EXPERTISE

Languages: Python, TypeScript, Java, Bash

Cloud & Containers: AWS, Docker, Kubernetes, Helm

CI/CD & Developer Tooling: Git, Bitbucket Pipelines, GitLab CI, GitHub Actions, release automation, security scanning

Python Tooling: uv, Ruff, ty, pytest, pre-commit — dependency management, linting, type checking, testing, and automated quality gates

Integration: REST/OpenAPI, RabbitMQ, ActiveMQ, distributed systems

Data: PostgreSQL, MySQL, SQL Server, Elasticsearch/OpenSearch, Redis